11C-choline vs. 18F-FDG PET/CT in assessing bone involvement in patients with multiple myeloma

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Background Multiple Myeloma (MM) is a B cell neoplasm causing lytic or osteopenic bone abnormalities. Whole body skeletal survey (WBSS), Magnetic resonance (MR) and 18F-FDG PET/CT are imaging techniques routinely used for the evaluation of bone involvement in MM patients. Aim As MM bone lesions may present low 18F-FDG uptake; the aim of this study was to assess the possible added value and limitations of 11C-Choline to that of 18F-FDG PET/CT in patients affected with MM. Methods Ten patients affected with MM underwent a standard 11C-Choline PET/CT and an 18F-FDG PET/CT within one week. The results of the two scans were compared in terms of number, sites and SUVmax of lesions. Results Four patients (40%) had a negative concordant 11C-Choline and 18F-FDG PET/CT scans. Two patients (20%) had a positive 11C-Choline and 18F-FDG PET/CT scans that identified the same number and sites of bone lesions. The remaining four patients (40%) had a positive 11C-Choline and 18F-FDG PET/CT scan, but the two exams identified different number of lesions. Choline showed a mean SUVmax of 5 while FDG showed a mean SUVmax of 3.8 (P = 0.042). Overall, 11C-Choline PET/CT scans detected 37 bone lesions and 18F-FDG PET/CT scans detected 22 bone lesions but the difference was not significant (P = 0.8). Conclusion According to these preliminary data, 11C-Choline PET/CT appears to be more sensitive than 18F-FDG PET/CT for the detection of bony myelomatous lesions. If these data are confirmed in larger series of patients, 11C-Choline may be considered a more appropriate functional imaging in association with MRI for MM bone staging.